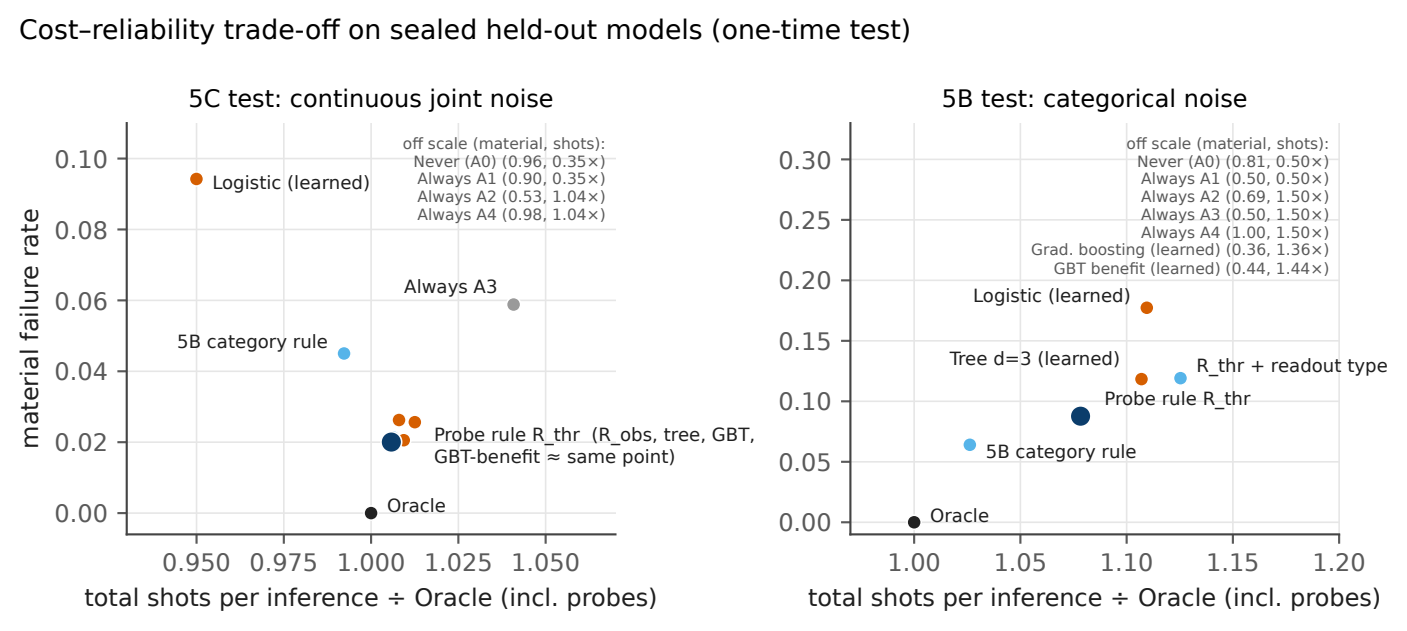
Cost–reliability trade-off on sealed held-out models (one-time test)
5C test: continuous joint noise
off scale (material, shots):
Never (A0) (0.96, 0.35×)
Always A1 (0.90, 0.35×)
Always A2 (0.53, 1.04×)
Always A4 (0.98, 1.04×)
Logistic (learned)
Always A3
5B category rule
Probe rule R_thr (R_obs, tree, GBT, GBT-benefit ≈ same point)
Oracle
material failure rate
0.10
0.08
0.06
0.04
0.02
0.00
0.950
0.975
1.000
1.025
1.050
total shots per inference ÷ Oracle (incl. probes)
5B test: categorical noise
off scale (material, shots):
Never (A0) (0.81, 0.50×)
Always A1 (0.50, 0.50×)
Always A2 (0.69, 1.50×)
Always A3 (0.50, 1.50×)
Always A4 (1.00, 1.50×)
Grad. boosting (learned) (0.36, 1.36×)
GBT benefit (learned) (0.44, 1.44×)
Logistic (learned)
Tree d=3 (learned)
R_thr + readout type
Probe rule R_thr
5B category rule
Oracle
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.00
1.05
1.10
1.15
1.20
total shots per inference ÷ Oracle (incl. probes)